

Supplementary material

The exact maximum growth factor for complete pivoting
on real 5×5 matrices

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Proof objects and reproducibility index

This supplement identifies the mathematical modules used in the article, their proof objects, and the scopes of the recorded exact acceptances. Historical labels beginning R, V, B, or Round identify archived sources; they are not additional mathematical hypotheses. File and collection names are manifest labels, not public URLs. Relative locations refer to the companion analytical collection or to the named extracted archive, as specified in S8. Hashes identify bytes; mathematical validity depends on the analytic arguments, exact acceptances, and source-composition rules. Large certificate payloads may be requested from the authors using the email address given in the article.

S1. The constant and attainment

`supplement/alpha_exact.json` contains all 62 ascending integer coefficients of the degree-61 polynomial and rational isolating endpoints $L < U$, with $U - L = 10^{-200}$. The published lower bound and its attaining equality family are due to Chen, Edelman, and Urschel. The identity record `evidence/v54/rho5_v54_alpha_id_entity.json` compares all coefficients and records the endpoint signs and the one-sign-change transformation $(1+t)^{61}P_{61}((4+5t)/(1+t))$. This identifies the same unique root in $(4, 5)$; it is not a separate global upper-bound proof.

The retained foundational reconstruction is in `supplement/foundations/`. R2 supplies coefficient and resultant reconstruction; R5 supplies rational Krawczyk isolation; R7 proves that the reconstructed matrix is actually completely pivoted and attains α . The readable attainment record is also copied to `supplement/R7_candidate_primal_cp.md`. Within the foundational directory the corresponding files are: `outputs/P5_coefficients_ascending.csv`, `outputs/P5_exact_verification.json`, `outputs/R2_free_CAS_reproduction_report.md`, `outputs/R5_candidate_interval_newton.md`, and `outputs/R7_candidate_primal_cp.md`. The actual entry points, preserving their original relative inputs, are `work/verify_p5_exact.py`, `work/candidate_interval_newton.py`, and `work/candidate_primal_cp_certificate.py`. They use `work/rational_interval.py` and the three polynomial files `work/sympy_recompute/p1.txt`, `work/sympy_recompute/p2.txt`, and `work/sympy_recompute/p3.txt`. The primal entry already rechecks the Krawczyk enclosure; running every entry separately is not required for every review.

S2. Early bounds, complete lift, and saturation

The scalar and saturation sources are `supplement/R8_scalar_root_bounds.md` and `supplement/R8_global_chart_coverage.md`. The complete two-step lift is R3, `supplement/foundations/outputs/R3_exact_lift_cheshev_reduction.md`. The early bounds apply after normalization of the largest initial entry to one. The D/X saturation used in the article preserves a nonzero fifth-pivot magnitude and the complete earlier pivoting bands; zero fifth pivots are handled by the early bounds.

The preserved historical R8 saturation text is accompanied by `supplement/R8_SATURATION_ERRATA_V55.md`. A zero fifth pivot need not mean a zero fourth-stage tail, and the two full-matrix constructions are asserted equal only at a positive-product junction. The immutable historical text has SHA-256

`0ffdc19be82c27f3e7d660bb915ce1bcc051908c89c246fba60cfe2b4ebecf0`.

The corrections, analytic proof map, and bounded checks are retained in `evidence/v55/ANALYTIC_AUDIT.md`, `evidence/v55/PROOF_DEPENDENCY_MAP.md`, and `evidence/v55/LOCAL_ACCEPTANCE_RECEIPT.json`. Those finite checks are not a replacement for the universal written arguments or a replay of the X and negative-D certificates.

S3. Canonical representative, reconstruction, and height

The canonical-height dependency has three distinct components. Their historical acceptance records must not be merged into a claim that the final composer re-executed all three.

Canonical representative (V24/T24). The original package is `RH05_V24_Exact_Verification_2026-09-06.zip`. Its six complete exclusion trees contain 228,542 nodes and 114,274 contradiction leaves, with no open leaf. The package retains the mathematical theorem, model, exact acceptor, and nested earlier inputs. Its SHA-256 is

`2478b3d6c8d9189fd508e2af4695bb2086c6628d5049c75cef759bd3e455dfa8`.

Here T24 denotes the canonical target proved in this package.

Row-scale reconstruction (V25). The original nested object is `RH05_V26_Exact_Verification_REBUILD/V25_DEPENDENCY.zip` inside `RH05_V26_Canonical_Global_Alpha_Exact_Verification_2026-09-07.zip`. The SHA-256 of the nested ZIP bytes is

`2ee17e8eb8ff5d21716675ce86eea2cc791dc2a9454d820c90bd481e164a47ca`.

The universal equivalence is analytic. Its 28 symbolic checks and rational regressions test implementation and reconstruction; they are not a sweep of all parameter values.

Canonical exact height (V28). The preserved package is `rho5_v28_exact.zip`, with SHA-256

`bec88339a158920d9dacb15341b02f359af83c4bf0e977721d9bdb2764fa0668`.

All thirteen localization fronts were continued. The proof has 152,467 records and 208,274 contraction certificates; its three terminal boxes in the peak neighborhood are paid by the analytic and algebraic height module. The independent replay used the earlier V24/V25 mathematical inputs without rerunning every older tree. V27 entrance dictionaries and other earlier dependencies are retained where the frozen component manifests require them.

S4. The complete cross-saturated domain

The complete X certificate collection comprises the following archive objects. The table gives their exact filenames and replay entry points; commands are run from the named extracted package root. Individual archive instructions specify their input dependencies.

Domain and source	Archive filename	Exact entry
$0 < k \leq 2$, V31	<code>rho5_v31_exact.zip</code>	<code>python verify_all.py</code>
$k \geq 2.2$, Round43	<code>ROUND43_MINIMAL_REPLAY.zip</code>	<code>bash repro/replay.sh, in RH05_CQG_ROUND43</code>
$2.15 \leq k \leq 2.2$, and strict high r for $2.1 \leq k \leq 2.2$, Round44	<code>ROUND44_MINIMAL_REPLAY.zip</code>	<code>python3 repro/replay_new.py .</code>

Domain and source	Archive filename	Exact entry
$2.14 \leq k \leq 2.15$, $r \leq k$, V33	rho5_v33_exact.zip	python verify_all.py
$2.1 \leq k \leq 2.14$, $r \leq k$, Round45	RH05_ROUND45_EXACT.zip	rho5_round45_exact/replay.py
$2 < k \leq 2.1$, $r > k$, corrected V34/Round46	RH05_V34_HIGH_EXACT_FULLL.zip	rho5_v34_high/replay_high.py
$\gamma/2 \leq k \leq 2.1$, $r \leq k$, $F \geq \gamma$, Round47	RH05_ROUND47_ALPHA_FULLL.zip	rho5_round47_alpha/replay.py

The last three ZIPs have respectively 222,151,979, 131,835,895, and 367,175,732 bytes. The last package, comprising complete roots I and II, has SHA-256

ddab67679b4faf3338beddaaea25dce99edfabeab937a0da07aa534c7a866de8.

Round46 uses the corrected exact factor-graph implementation, not the superseded one. V36 assembles the unrestricted complete X theorem: `supplement/V36_THEOREMS.md`, with its full source at `rho5_v36_exact/THEOREMS.md`. The package instructions specify Python/SymPy, C++17, and Boost requirements. Their previously recorded acceptances are not represented as new runs performed in preparing this index.

S5. Six-variable maximum in the negative diagonal domain

V36 gives the attained actual-tail maximum and qualified double-gap flow. V37 eliminates (β, p, e, r, s, t) at a fixed seventeen-coordinate framework, on each nonempty high-value fibre in the normalized $p \geq 1$ range. These are same-source feasibility statements, without packet-rank assumptions. The preserved texts are `supplement/V36_THEOREMS.md` and `supplement/V37_SIX_VARIABLE_ELIMINATION.md`; the full V37 source is `rho5_v37_exact/theory/SIX_VARIABLE_ELIMINATION.md`. The complete framework model is `supplement/B17_FULLL.json`, SHA-256

39a65f2cbd6089dab3a403c6a72f365a02003737a641be86c3cf73cb70e76de3.

The model contains canonical images of every hypothetical source above α , including the proper negative domain with $k \leq 2$. It is not restricted globally to a selected contact equation.

S6. Acceptance rules and their thresholds

Let $\gamma = 4132517/10^6 < \alpha$. Contradiction rules exclude complete sources with $F \geq \gamma$; height and qualified transport rules may instead establish only $F \leq \alpha$. An alpha-safe source need not be infeasible at the lower threshold. Exact nonnegative-weight, interval, contraction, and ancestor-image checks are part of acceptance, not merely discovery output or hash comparison.

The final local record has 47 old contradiction leaves, three alpha-safe leaves (H, A, B14), one V53 contradiction, one B15 canonical-source threshold exclusion, and two B16 canonical-source threshold exclusions. Each B15/B16 source covers both qualified conditional charts. The joint conclusion is alpha safety, not whole-anchor gamma infeasibility.

Within the complete B preservation archive, the actual mathematical interfaces include `deep_math.py`, `alpha_cover.py`, and the frozen depth500 verifier; source assembly uses `alpha_parent_cover.py` and `final_overlay_accept.py`. The final local mathematical producer `verify_joint_anchor.py` has SHA-256

548de0119a5d69ff543ab4da4c5c7c5fd6a63e7f5707c4c5c70a7af4ad548bba.

The source composer `final_overlay_accept.py` has SHA-256

974b79b97732b02008207fec847a8a68796fcb939c5acf862340cb6c5acc0a13.

The joint producer executes its frozen child verifiers; the later composer consumes identified earlier mathematical receipts and checks their source bindings. It does not re-execute all inherited mathematics.

S7. Complete negative-diagonal coverage and binding

The preserved base tree is `rho5_independent_acceptance_20260911_11/checkpoint/B17_ROOT.json` inside the historical-root archive. It has 2,102,766,386 bytes and SHA-256

8e138dce9a65e49252a1f4714229b1ca95227943913f1eeefdf51714810a563c.

Its exact cold replay accepted 749,693 nodes: 374,846 splits, 374,839 contradiction terminals, four alpha-safe terminals, and four explicit old-rule open terminals. The 750 enclosed B44 continuations have no internal open terminal. The cold record `rho5_independent_acceptance_20260911_11/COLD_REPLAY.json` has SHA-256

55f3a9dd5090cfd9e21c479444d5cca1fc15fde2b5c22b8180bbd1c834156b37.

The recorded Fraction replay used eight workers and took 22,815.35 seconds. That is a historical mathematical acceptance, not the runtime of the final composition.

The article's Q_1, Q_2, Q_3, Q_4 are the following actual old-root boxes; the historical indices are retained here to locate their payloads:

Article	Index	Original-root path	Paid responsibilities
Q_1	338726	10011000010101010	111/111
Q_2	435003	10111000001100011000010111011	240/240
Q_3	563285	1101000101111	159/159
Q_4	675104	1110101001111	56/56

The last local subtree, at relative path 10011101, covers 54 terminal positions and pays only the final one of the 240 responsibilities of parent 435003. The original-parent accounting is 465 old accepted parents plus these four completions, hence 469/469 and effective open count zero. The old root bytes still contain four O records; they are paid by the separate cover, not relabelled. The article combines the historical types from S6 into 48 contradiction terminals, three height or transport terminals, and three canonical-source threshold exclusions. These are the same 54 terminal responsibilities, not additional proof objects.

Final local proof. `evidence/joint/ANCHOR_COMPLETE.json` and `evidence/joint/COMPLETED_ANCHOR_LEDGER.json` record the B15/B16/V53 joint acceptance. The former, from the recorded 184.37-second run, has SHA-256

648aa5322cfd518d1eaa549bf806b3daab67123078c2aa64909ba97831dd11d1.

Last parent. `evidence/final_overlay/PARENT_435003_COMPLETE.json` has SHA-256

6451a87db7383915b58bb3174581a4c81c38eca9a8aae32641d00083ad6cacb9.

Actual root cover. `evidence/final_overlay/ROOT_COMPLETE_OVERLAY.json` has SHA-256

43dbb784c871ecad4b6ff10047f39a2dad7a69a0b6eb392b40c9a56306e6ea34.

Its 357 direct dependencies are enumerated in `FINAL_OVERLAY_DIRECT_DEPENDENCIES.json`. Recursive runtime and upstream theorem dependencies are additional objects; this list alone is not the complete proof archive.

S8. Artifact collections and reproducibility scope

The manifest groups the proof objects into the following logical collections. These names identify sets of retained files, not public repositories or newly assigned archive identifiers.

Companion analytical collection. The relative `supplement/` and `evidence/` locations used in S1–S7 contain the analytical sources, selected exact inputs, and acceptance records. The foundational `work/` and `outputs/` inputs retain their relative structure under `supplement/foundations/`; their identities are listed in `FOUNDATIONAL_MINIMUM_MANIFEST.json`.

Canonical and complete X certificates. The archive objects named in S3–S4 include their frozen mathematical inputs and acceptors. Their manifests determine the nested dependencies.

Complete negative-diagonal certificates. The historical-root and final-cover archives below contain the base root, its exact runtime, and the four complete source compositions.

The two complete negative-diagonal archives are:

historical_root.tar.gz 2,189,588,699 bytes; SHA-256 15ef795e85d0d295e92fdd629f313b9c284cc6519388011862aee3c1e18e3f73.

final_cover.tar.gz 1,385,430,364 bytes; SHA-256 1c9fb99d184eaf9d79b44d33826e4db8fbced8db41ee2b46ec2e1616bd12361c.

Their retained copies have passed whole-archive hash checks, recorded in the preservation collection's `LOCAL_ARCHIVE_RECEIPT.json`. This verifies byte identity, not a further mathematical replay. The second archive retains the relative locations `rho5_deep500_all_open_20260910_08/`, `rho5_b16_v53_closure_20260911_18/`, `rho5_b16_20260911_17/`, and `rho5_cqg_continuous_20260910/run_01/runtime/models/B17_FULL.json`. The first contains the base tree and its `runtime/R54_SOURCE_MANIFEST.json`, listing 959 runtime sources, under `rho5_independent_acceptance_20260911_11/`. These are archive-relative locations. Frozen absolute-location bindings require an explicit replay mapping that retains every mathematical input identity and source check.

Final local source. The 18,292,611-byte final-anchor review bundle, SHA-256 88a05ffe78a54d0c552aa6c5f2294e8614b7f78296efbbfb0cd5332322d21e15, contains a standard-library-only short replay for Python 3.10 or later:

```
python -S restore_inputs.py
python -S verify_joint_anchor.py --v53-dir rho5_v53_exact \
    --b16-dir B_STRUCTURE_16_DELIVERY --output-dir replayed_anchor
```

The output directory must be empty or absent. The result pays the 54-leaf anchor only; its standalone whole-parent and whole-domain flags correctly remain false.

Complete negative-diagonal domain. From the extracted historical-root directory, the exact root entry is:

```
python -B -S runtime/r54_replay.py --tree checkpoint/B17_ROOT.json \
    --out REPLAY_ROOT_NEW.json --workers 8 --backend fraction --allow-open
```

The new output must not already exist. Its four expected O records are covered by the separately accepted parent proofs. `deep_math.verify_branch` and `alpha_cover.verify_cut` are mathematical interfaces; `alpha_parent_cover.py` and `final_overlay_accept.py` compose accepted sources without rerunning all the mathematics of their input receipts. A complete independent replay therefore includes both the parent mathematics and the final source binding. The frozen 239/240 baseline must be preserved when attaching the last local proof, with a hash-bound replay configuration and fresh output locations. Python assertion optimization must not be enabled.

Full real-matrix theorem. The global conclusion also uses S1–S5, including the complete X certificates, attained lower bound, and analytic quantifier and boundary arguments. The recorded computation consists of exact component acceptances and verified compositions. It is not a new single-command fresh-environment replay of the entire dependency chain. `REPRODUCIBILITY.md` specifies the recorded entry points and their scopes; `THEOREM_CHAIN.md` gives the dependency map. The identities and acceptance scopes in this supplement distinguish preservation, mathematical replay, and composition. They do not assert an existing public deposition or a newly completed independent replay of every historical component.